JANGARA BLISS

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EDUCATION

Fort Lewis College (FLC), Durango, CO — B.S. Computer Engineering Expected May 2027

Minors in Mathematics and Business Administration · Cumulative GPA 3.65 · Upper-division GPA 3.80

Coursework: Intro to Robotics; Systems & Control; CE 433 Embedded Devices (FPGA design); Digital Signal Processing; Engineering Statistics (calculus-based); Differential Equations; Calculus III. Fall 2026: Linear Algebra, Discrete Math, Analog Electronics; Spring 2027: semiconductor fab (FLC nanolab).

Embedded projects: custom ATmega PCB design; FLC microcontroller-board bring-up with C drivers (GPIO, PWM, ADC, UART).

RESEARCH & ENGINEERING EXPERIENCE

Research Assistant, Humanoid Robot Intelligence May 2026 – Aug 2026

Advisor: Dr. Yiyan Li · Booster K1 humanoid

- Built a two-tier stack: NaVILA, an 8-billion-parameter vision-language navigation model (~1 Hz), commanding a reinforcement-learning locomotion policy I trained in Isaac Sim/Isaac Lab (50 Hz; simulation only).
- Integrated the K1 into the VLN-CE-Isaac benchmark and ran all **1,077** episodes; over 22 training runs (~270 GPU-hours) and pipeline fixes, including termination and camera-warmup fixes that more than doubled walking distance, raised success from under 1% to **18.3%**.
- Ran a preregistered **2,400-episode** ablation: two identically configured 300-episode runs agreed in aggregate (15.0% vs. 16.0%) but disagreed on **11.0%** of episodes (Cohen's kappa = 0.58); a fixed time limit censored conditions unequally (2.0% vs. 12.3% timeouts).
- Built a relay (RTX 5090 VLM inference, ~350 ms; Booster SDK velocity control) that ran NaVILA live on the physical K1 for two natural-language episodes with zero teleoperation, using Booster's built-in walker.
- Built a ROS 2 cart-seek stack (YOLO-World): walks the K1 to a cart, stops it ~1 m away (Booster's walker); ~25 live runs.

Senior Design: Bridge Crack-Inspection Robot Arm Sep 2026 – present

Advisor: Dr. Yiyan Li · reBot B601-DM, 6+1 DoF

- Brought up all seven motors over CAN with a **200 Hz** control loop (setpoint, stiffness, damping, feed-forward torque per motor); built the Jacobian, inverse kinematics, and a minimum-jerk planner, testing every move in simulation first.
- Measured the elbow's gravity load with an up-and-down torque test (6.6 N·m vs. 2.7 N·m in the URDF); gravity/friction feed-forward and gain tuning cut loaded-joint holding error from 0.04–0.05 to at most **0.005 rad** (41 logged runs).

Research Assistant, Computer Vision & Robotics May 2025 – Sep 2025

Advisor: Dr. Kevin Wedeward · Solo project

- Revived two non-functional Rethink robots (Sawyer, Baxter) with no vendor support: traced Sawyer's boot failure to a dead CMOS battery that had reset its BIOS into an incompatible Secure Boot setting, got the needed Intera update through outreach to other universities, and set up ROS 1/ROS 2 with Docker, MoveIt, and Gazebo.
- Trained a YOLOv8 solar-panel thermal hotspot detector with a Blender synthetic-data pipeline and checked it qualitatively on real panels; **2nd place**, FLC Physics & Engineering Symposium, the highest-placing solo project.

Team Lead, Colorado Robotics Challenge (NASA Space Grant) Oct 2025 – Apr 2026

Advisor: Dr. Charles Hakes

- Led a 4-person team; designed the electrical system of an 18-DoF hexapod (Arduino Mega, 9-DoF IMU, bump sensors, LiPo + SBEC power) and extended its firmware: gyro-magnetometer heading hold, gyro-bias calibration, bump avoidance.
- Built a Kuramoto coupled-oscillator gait simulator (tripod, wave, ripple gaits from one system); open-sourced, **34-test** CI.

Research Assistant, AI & Data Aug 2024 – Oct 2025

Advisor: Dr. Matthew Welz

- Built a Power Apps music-library database with the FLC radio station, designed for 60+ daily users (handed off pre-deployment); prototyped embedding-based artist-name matching and LLM agents querying a Neo4j graph in natural language.

AWARDS, HONORS & PROGRAMS

2nd Place, FLC Physics & Engineering Symposium (2025) · Katz School of Business Leadership Award (2024) · New Registered Student Organization of the Year (2023–24) · Goldman Sachs Emerging Leaders Series (2024)

LEADERSHIP & SERVICE

- **Co-Founder & President, Entrepreneurial Ventures Association** (2023–24): led an 8-person executive team; ran a pitch competition allocating $1,500 in micro-grants; brought NASA's Technology Transfer University (T2U) program to campus.
- **Student Representative, FLC Strategic Implementation Committee** (2025–26): one of four students working with the college president, trustees, and deans on the 2025–2030 Strategic Plan.
- STEM Tutor (2024–25) · Chip Competition Manager, Nicholas Cisewski Memorial Golf Tournament for pediatric cancer research (2020–present)

SKILLS

Robotics & ML: PyTorch, reinforcement learning, Isaac Sim/Isaac Lab, MuJoCo, ROS 1/ROS 2, MoveIt, Gazebo, YOLOv8, YOLO-World, Open3D

Control & Hardware: PD and feed-forward motor control over CAN, trajectory planning, board bring-up, soldering, embedded C drivers (GPIO, PWM, ADC, UART), ATmega board design, oscilloscope, logic analyzer, FPGA (Vivado)

Programming: Python, C++, C, Verilog, SQL; Linux, Docker, Git · Spanish (conversational)